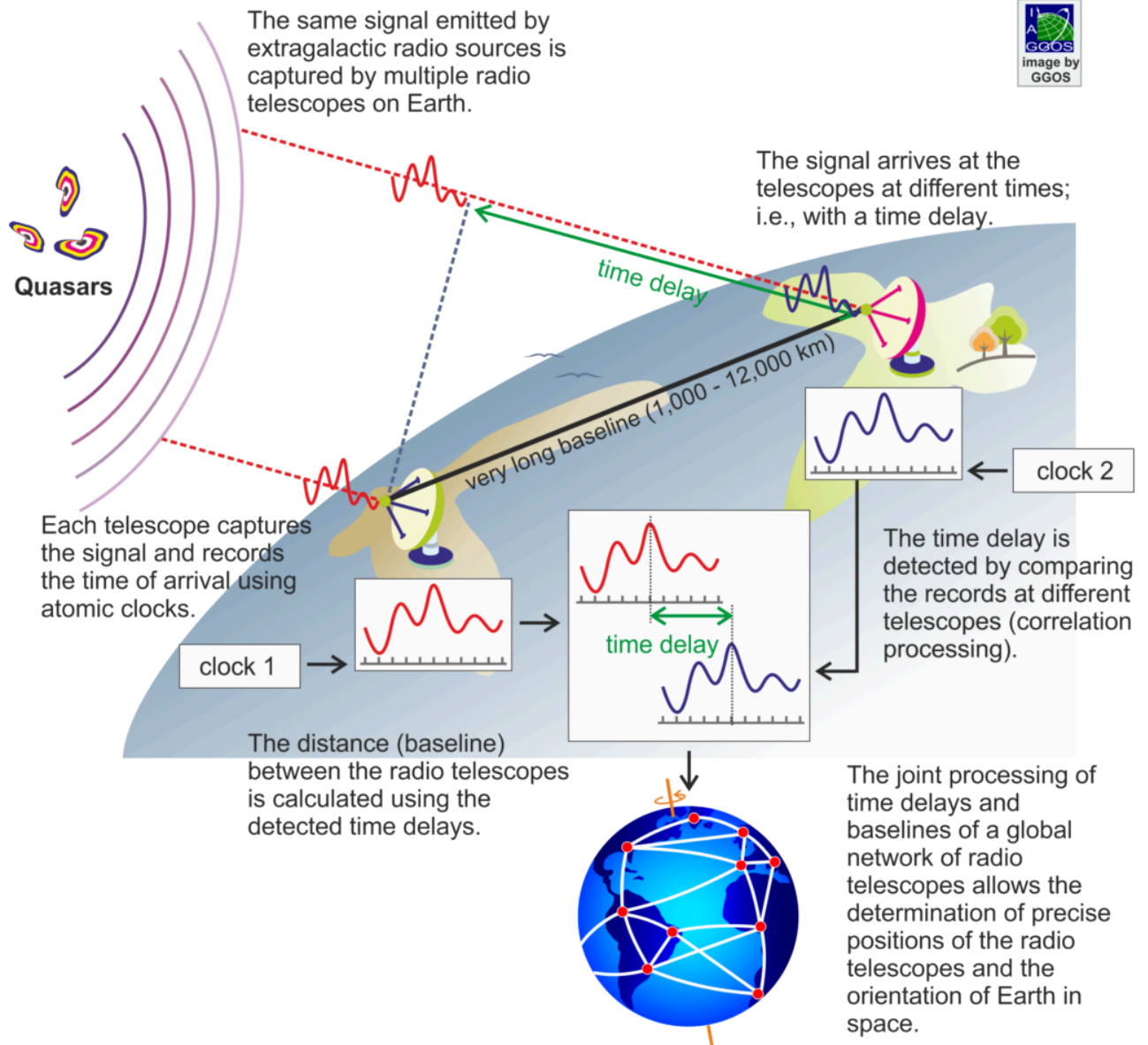


L3: International Terrestrial Reference System

- Geodetic reference systems required as an absolute reference
- Geodetic reference systems as a metrological basis
- **Terrestrial reference system**
 - Cartesian coordinates
 - Earth-fixed
 - Co-rotating with Earth
 - Defining:
 - Origin: Geocenter
 - Scale: SI meter
 - Orientation
- ITRF2020 latest ITRF
 - **Combination** of DORIS, GNSS, SLR, VLBI measurements
- VLBI



o

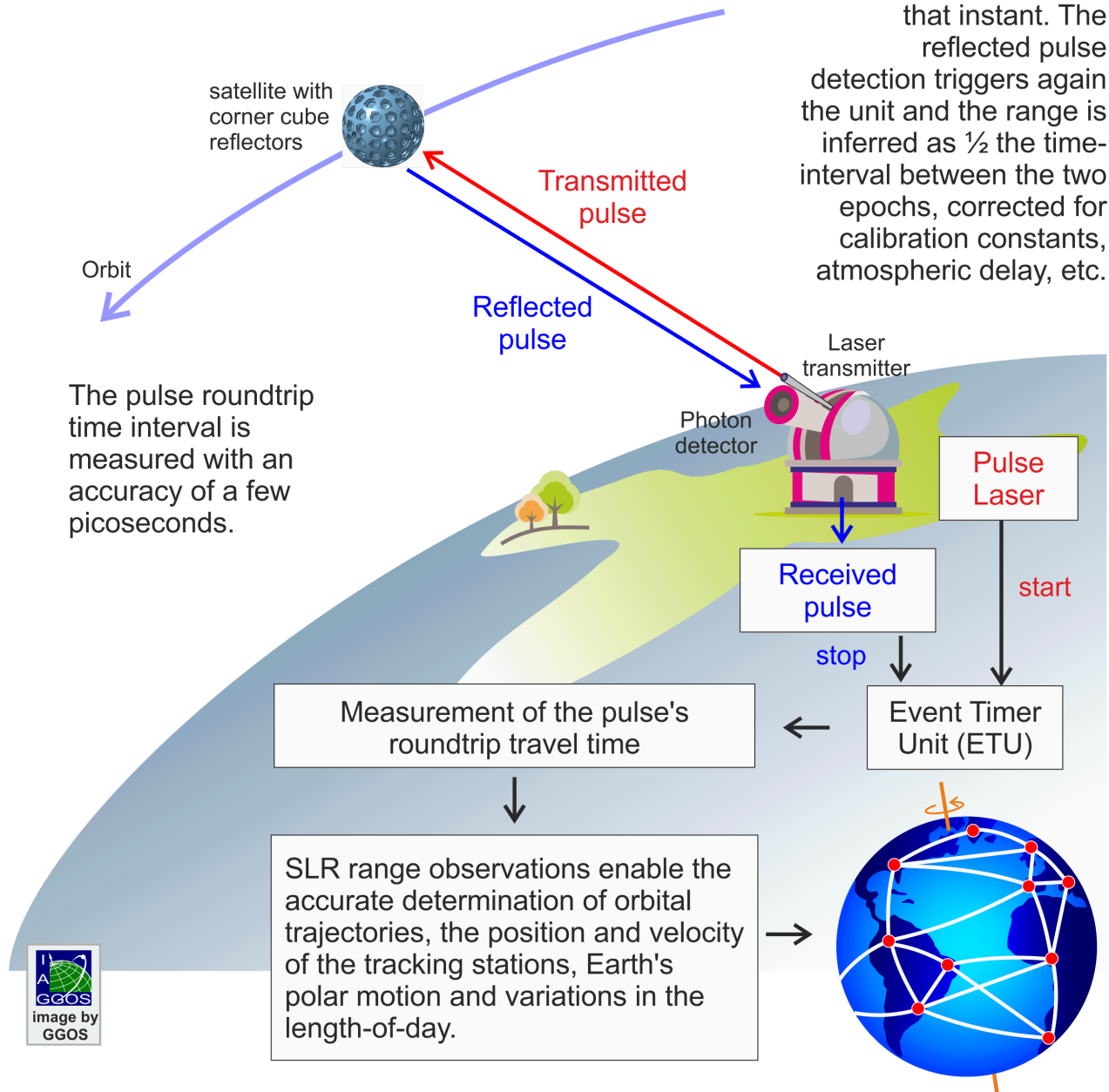
o Importance:

- For ITRF scale
- ICRF - extragalactic radio sources
- 5 EOPs

- SLR

Earthbound observatories emit ultra-short laser pulses towards retroreflector arrays on satellites or the moon surface, which reflect the pulses back towards their origin.

The emission of the pulse triggers an Event Timer Unit that records the precise epoch of that instant. The reflected pulse detection triggers again the unit and the range is inferred as $\frac{1}{2}$ the time-interval between the two epochs, corrected for calibration constants, atmospheric delay, etc.



o

o Origin of TRF

o With VLBI -> Scale

• GNSS

o Applications for high-precision Earth observation

• DORIS

o Precise orbit determination of Low Earth Orbiting (LEO) satellites

o Small and low cost

o Best station network w.r.t. global and homogeneous distribution

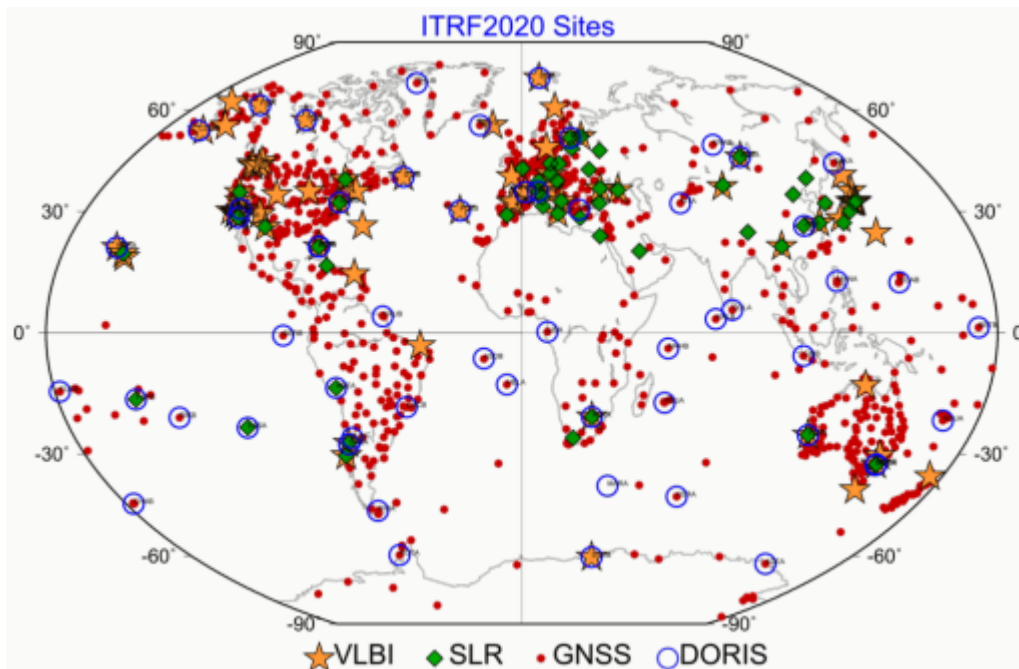
Current Combination Strategy

• Combination of 4 space geodetic techniques via local measurements on the ground:

- **Local ties**

- Co-located sites to connect different techniques

- ITRF Network



International Terrestrial Reference Frame (ITRF)

- Augmented Parametric Reference Frame (since ITRF2020):

$$\mathbf{X}(t) = \mathbf{X}_0(t_0) + \dot{\mathbf{X}}(t - t_0) + \delta\mathbf{X}_{PSD}(t) + \delta\mathbf{X}_S(t)$$

- GNSS station Potsdam POTS 14106M003, $t_0 = 01.01.2015$, [July 18, 2018; end]

$$\mathbf{X}_0(t_0) = \begin{pmatrix} 3800689.4801 \\ 882077.5455 \\ 5028791.4169 \end{pmatrix} \text{ m}$$

$$\dot{\mathbf{X}} = \begin{pmatrix} -0.01591 \\ 0.01606 \\ 0.00944 \end{pmatrix} \text{ m/a}$$



GFZ Potsdam, A17

Realisations

- 3 combination centers of IERS
 - Deutsches Geodätisches Forschungsinstitut (DGFI-TUM)
 - DTRF2020
 - Institut National de l'Information Géographique et Forestière (IGN)
 - ITRF2020
- Jet Propulsion Laboratory (JPL)
 - JTRF2020

- Responsible to provide TRF products by combining ITRF inputs from the Technique Centres and others
- Regional Densifications for each Continent

State-Of-The-Art

- Geodetic reference frame as metrological basis for Earth system monitoring
 - Requirements 1mm - SotA: 5mm origin, 1mm scale
 - Long-term stability - SotA: 1mm/dec - 5mm/dec
- Goals by Global Geodetic Observing System
- SotA: ITRF2020

Summary

- Reference frame as the metrological basis for Earth system monitoring
- Combination of GNSS, SLR, VLBI, and DORIS to determine accurate and stable global terrestrial reference frames
- Current realizations: ITRF2020, DTRF2020, JTRF2020
- Current accuracy does not fulfill required accuracy → Limitations
- Investigation of technique-specific systematics and combination strategy essential (current research)